

Beyond Boxes

Dense, Structured, and Promptable Vision

What should a vision model output when a box is not enough?

One image.

Several kinds of answer.



| The question determines the structure of the answer.

One image, many questions



- class
- boxes
- pixel labels
- instance masks
- keypoints
- depth
- prompted region

One image, six questions



What is present?

class / classes

Where are the objects?

boxes

What class is every pixel?

semantic field

Which pixels form each person?

instance masks

Where are body joints?

keypoints

How far is each surface?

depth

Same input. Different target structure.



SAME IMAGE

• class

• boxes

• pixel labels

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• depth

• prompted region

Label every pixel



- class
- boxes
- **pixel labels**
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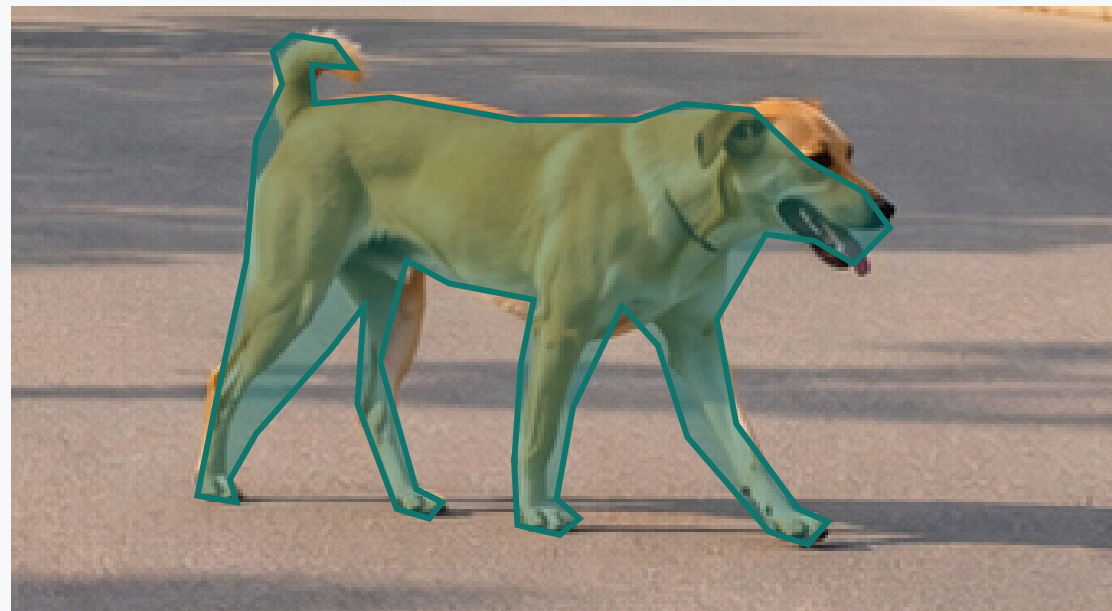
A box does not describe the silhouette

Where is the dog?



A rectangle also contains road and empty space.

Which pixels belong to it?



A mask describes the visible shape.

A box gives approximate extent. Segmentation specifies pixel membership.



SAME IMAGE

• class

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• prompted region

Semantic segmentation labels the image grid



- sky
- road
- person
- dog
- car

Selected regions shown.

Each valid pixel receives one class label under the chosen annotation scheme.



SAME IMAGE

— • class

• boxes

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One pixel has one target and C competing scores



Which class is this pixel?

0	1	1	0
1	1	1	0
1	1	0	0
2	2	2	2

Target: person

$$y_{ij} = \text{person}$$

pixel feature

$$f_{ij} \in \mathbb{R}^d$$

sky -2.303

person -0.357

road -1.609

C class logits

The network produces scores; supervision supplies the correct class.



SAME IMAGE

— • class

• boxes

• **pixel labels**

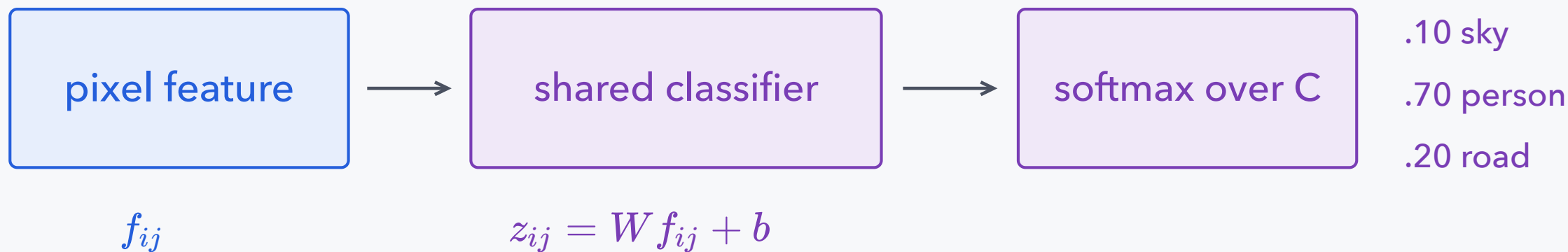
• instance masks

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The pixel classifier shares its weights



$$\sum_{c=1}^C p_{ij}(c) = 1$$

One probability distribution per pixel.

Softmax normalizes over classes separately at each location.



SAME IMAGE

—• class

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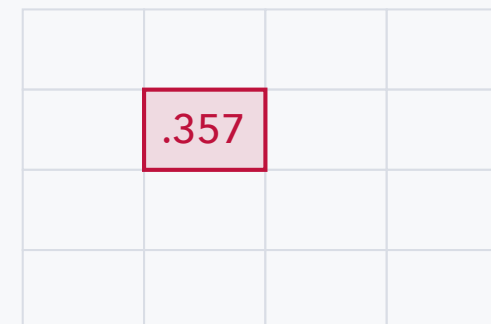
Cross-entropy asks for the target class probability

Target: person

sky		0.10
person		0.70
road		0.20

$$\ell_{ij} = -\log(0.70) = 0.357$$

Use the target class probability.



Place this loss back at the selected pixel.

A confident mistake receives a larger penalty than a correct high-probability answer.



SAME IMAGE

—• class

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The image loss averages valid pixel losses

0.08	0.12	0.21	0.06
0.18	0.357	0.09	✕
0.17	0.23	0.14	0.05
0.11	0.08	0.32	0.1

$$\mathcal{L} = \frac{1}{|V|} \sum_{(i,j) \in V} \ell_{ij}$$

$$\ell_{ij} = -\log p_{ij}(y_{ij})$$

One loss per labelled pixel

Crossed cell: ignored

V contains valid annotated pixels.

Semantic segmentation applies shared classification over many spatial locations.



SAME IMAGE

→ class

• boxes

• pixel labels

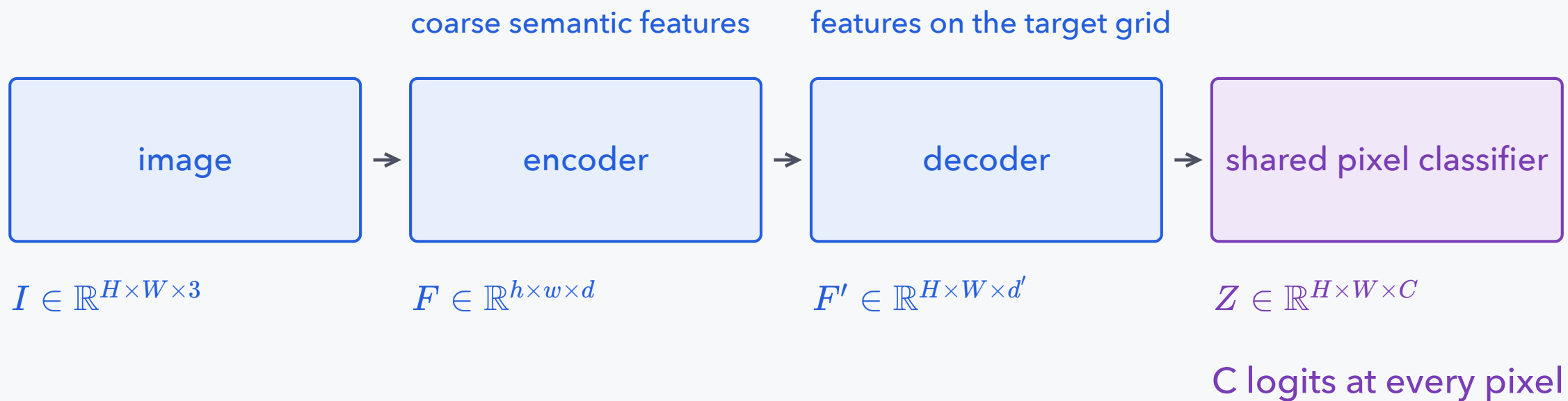
• instance masks

• keypoints

• depth

• prompted region

A spatial pipeline produces one classifier input per pixel



The encoder supplies context; the decoder returns features to the target grid.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

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How do we recover spatial detail?

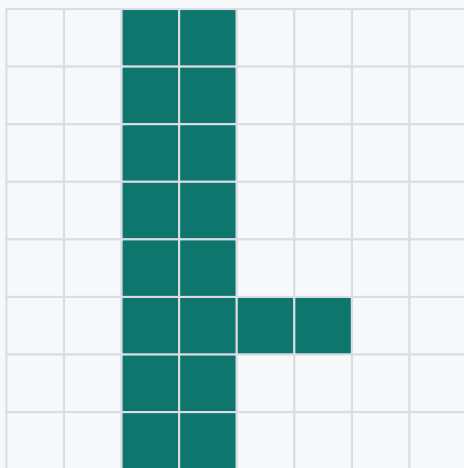


- class
- boxes
- **pixel labels**
- instance masks
- keypoints
- depth
- prompted region

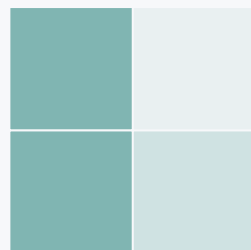
Upsampling increases resolution, not evidence

image → encoder → decoder → pixel classifier

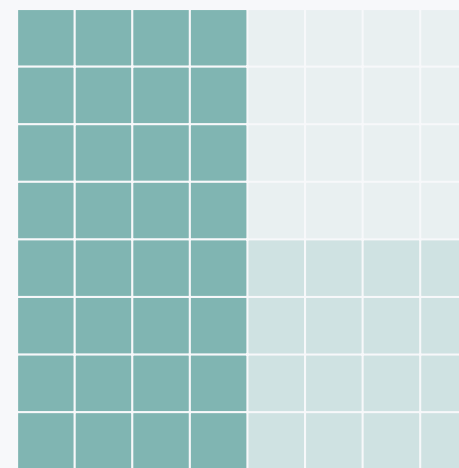
Fine mask



Downsample



Upsample



Is a larger grid enough?

The four coarse values do not identify the original boundary.

Several different fine boundaries can produce the same coarse summary.



SAME IMAGE

→ class

• boxes

• pixel labels

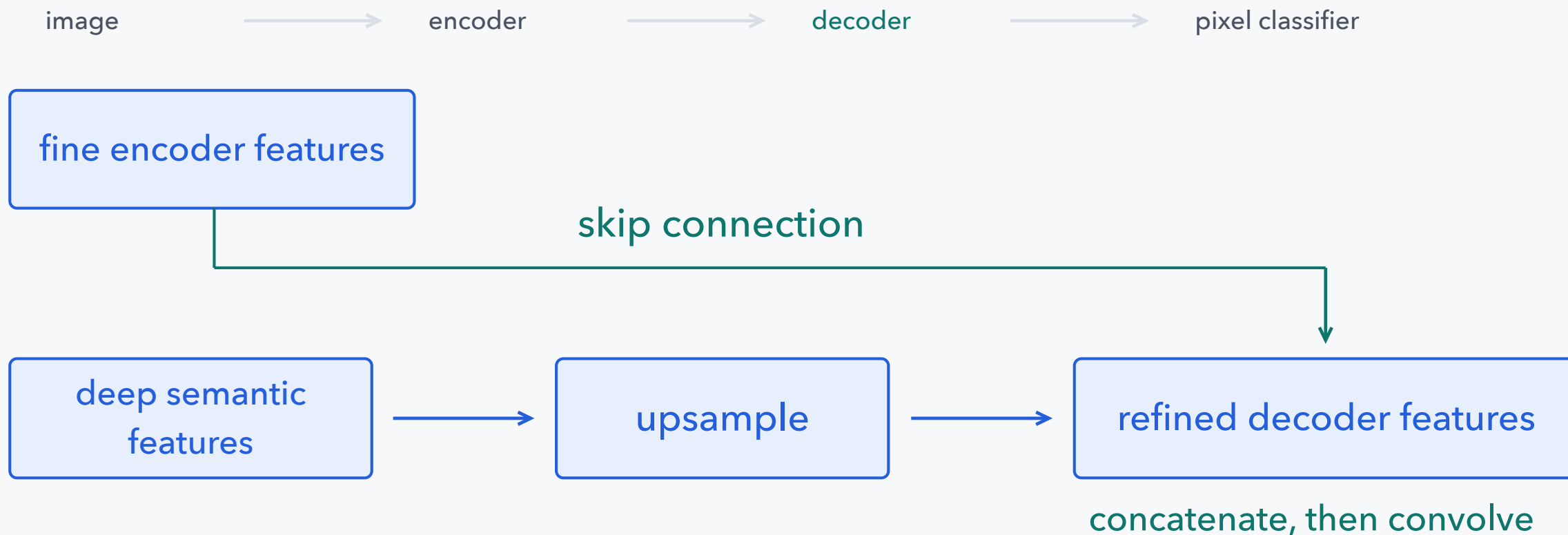
• instance masks

• keypoints

• depth

• prompted region

Where could the missing detail come from?



A skip connection brings fine encoder features into the decoder.



SAME IMAGE

• class

• boxes

• **pixel labels**

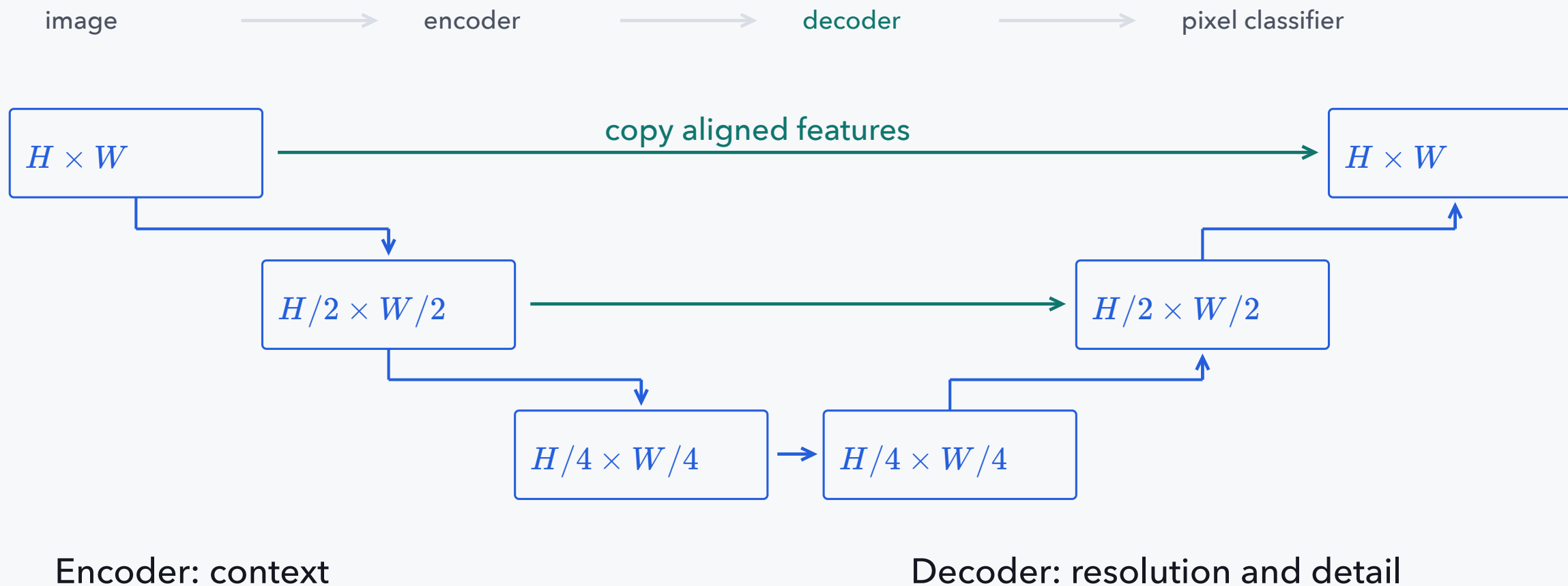
• instance masks

• keypoints

• depth

• prompted region

U-Net repeats this combination across scales



At each decoder scale, aligned encoder features help refine the prediction.



SAME IMAGE

• class

• boxes

• **pixel labels**

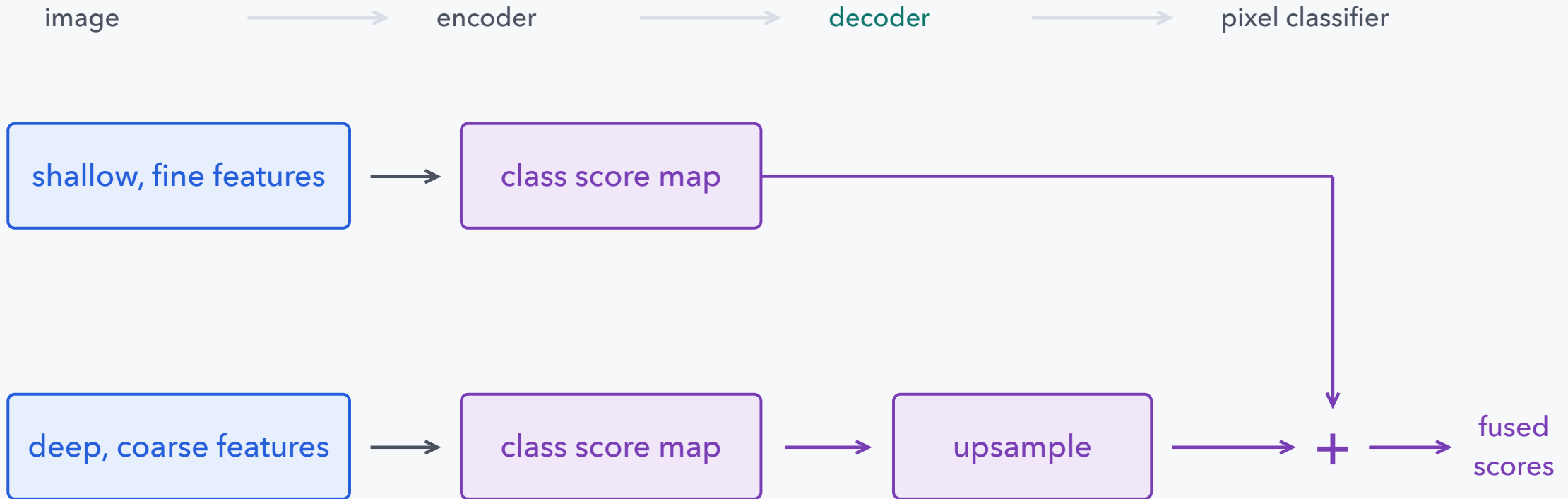
• instance masks

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FCN also combines coarse and fine predictions



FCN fuses score maps. U-Net concatenates features before further convolutions.

The common problem is precise localization from a hierarchy of features.



SAME IMAGE

→ class

• boxes

• **pixel labels**

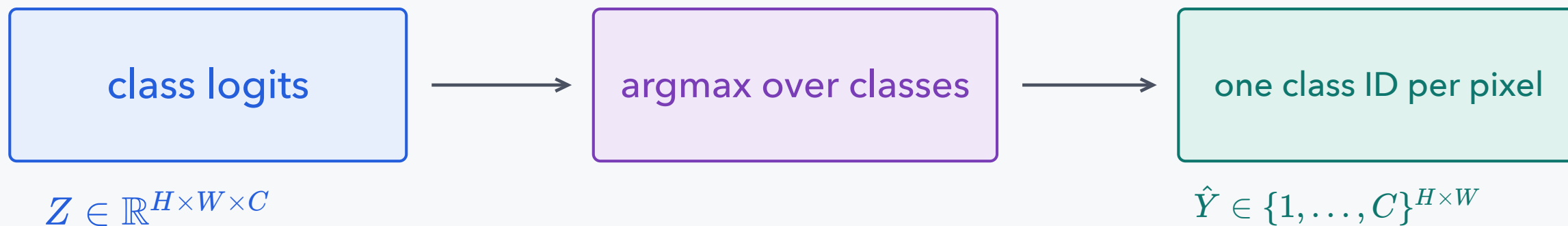
• instance masks

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• depth

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Inference turns class scores into a label map



$[0.10, 0.70, 0.20] \longrightarrow \text{person}$

The decoded output has one class ID per pixel.



SAME IMAGE

—• class

• boxes

• **pixel labels**

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Mask IoU measures foreground agreement

Target G

0	1	1	0
0	1	1	0
0	1	1	0
0	0	0	0

Prediction P

0	1	1	0
1	1	1	0
0	1	0	1
0	1	0	0

Overlap and errors

	TP	TP	
FP	TP	TP	
	TP	FN	FP
	FP		

TP = 5

FP = 3

FN = 1

$$\text{IoU} = \frac{TP}{TP+FP+FN} = \frac{5}{9} \approx 0.556$$

Five pixels agree on foreground; four more belong to only one mask.



SAME IMAGE

—• class

• boxes

• pixel labels

• instance masks

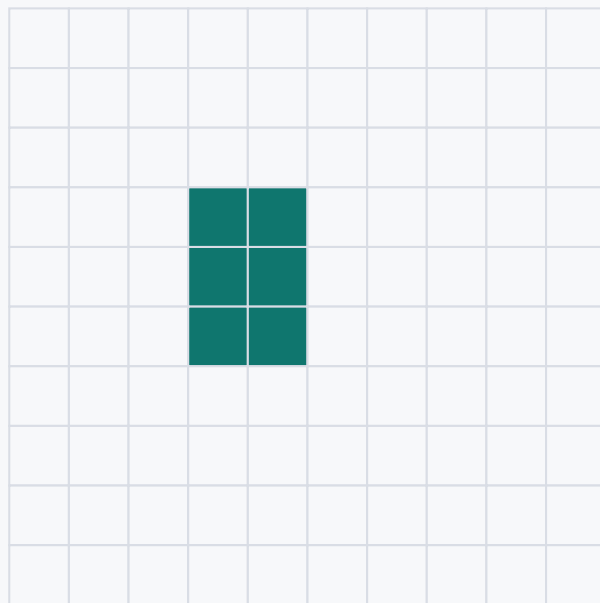
• keypoints

• depth

• prompted region

Background can dominate pixel accuracy

6 foreground pixels



predict background
everywhere



94% accuracy

Foreground IoU = 0

Foreground overlap and class-wise IoU expose this failure.

A high accuracy can coexist with complete failure on a small foreground region.



SAME IMAGE

— • class

• boxes

• **pixel labels**

• instance masks

• keypoints

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Which pixels belong to which instance?



- class
- boxes
- pixel labels
- **instance masks**
- keypoints
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A semantic class can contain several individuals

All three have class person

Each person also has an instance ID



Which class is this pixel? Which object does this pixel belong to?

Semantic labels give the class. Instance IDs give the individual.



SAME IMAGE

— • class

• boxes

• pixel labels

• **instance masks** • keypoints

• depth

• prompted region

Each instance has its own binary mask



Recall one detection record

$$(c_i, b_i)$$



$$\{(c_i, b_i, M_i)\}_{i=1}^N$$

$$M_i \in \{0, 1\}^{H \times W}$$

N varies by image. The object records are unordered.

The new field in the familiar object record is the binary mask.



SAME IMAGE

— • class

• boxes

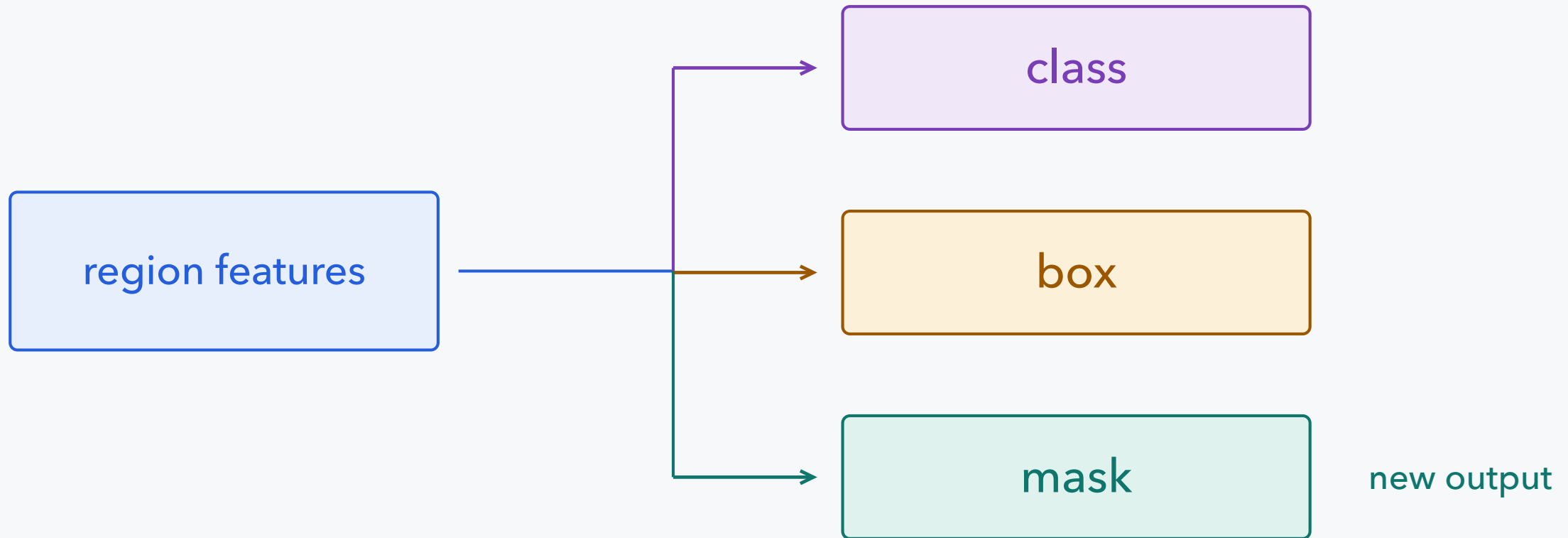
• pixel labels

• **instance masks** • keypoints

• depth

• prompted region

What extra detector output would describe shape?



A mask branch extends the detector's object record.



SAME IMAGE

• class

• boxes

• pixel labels

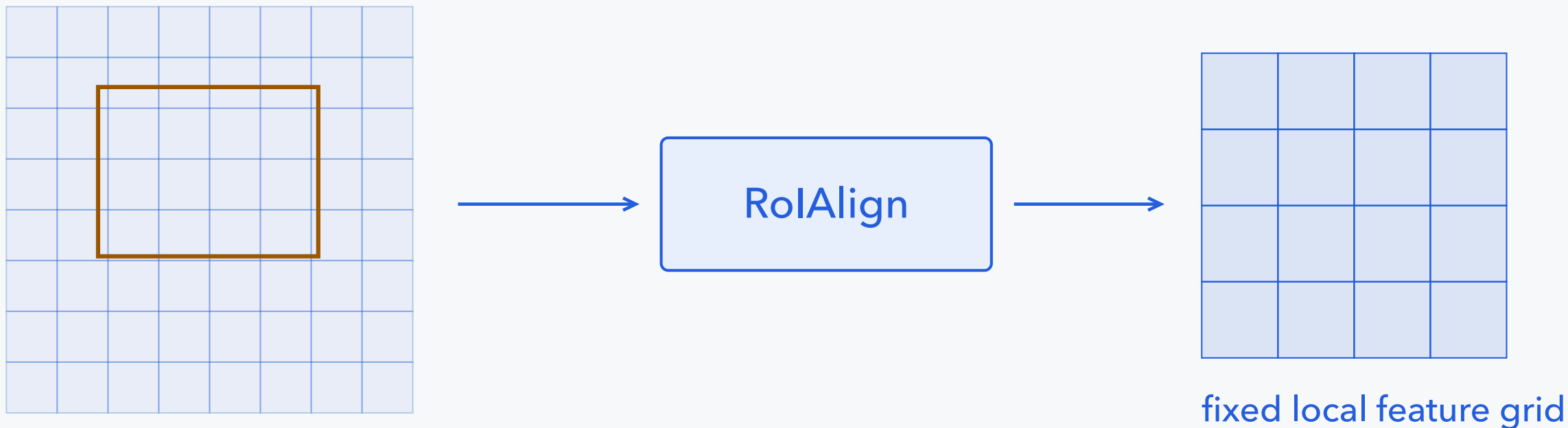
• **instance masks** • keypoints

• depth

• prompted region

A detector can provide features for one region

Image feature map + proposal



Sample aligned features; preserve the spatial layout within the region.

A region feature grid preserves spatial layout inside a proposed object.



SAME IMAGE

— • class

• boxes

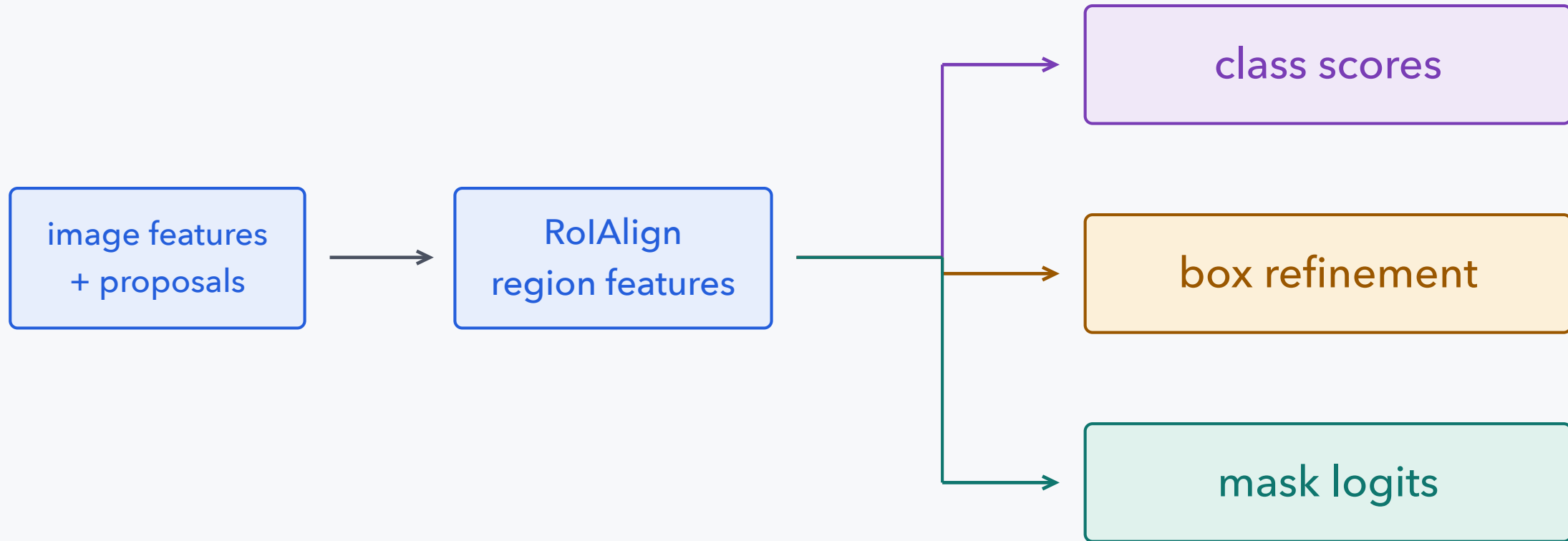
• pixel labels

• **instance masks** • keypoints

• depth

• prompted region

Mask R-CNN adds a mask branch to region detection



Shared region features support classification, box refinement, and a local mask.



SAME IMAGE

— • class

• boxes

• pixel labels

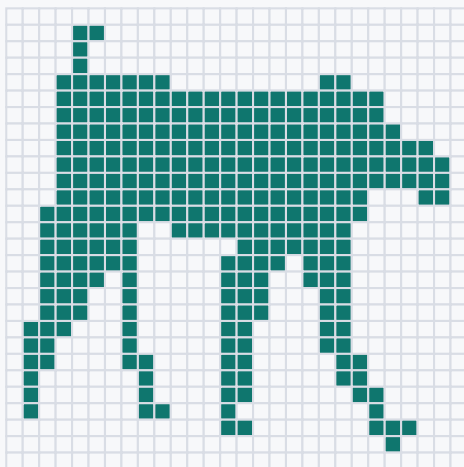
• **instance masks** • keypoints

• depth

• prompted region

A local mask is placed back into the image

Local binary mask



28×28



resize to box



place in full image

The mask grid predicts shape. The box supplies image location.



SAME IMAGE

— • class

• boxes

• pixel labels

• **instance masks** • keypoints

• depth

• prompted region

The mask loss learns binary membership

A matched positive region has three supervised outputs.

class

box

binary mask

$$\mathcal{L} = \mathcal{L}_{\text{class}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}}$$

The mask term learns foreground membership within the matched region.

The mask branch answers foreground or background within the matched region.



SAME IMAGE

→ class

• boxes

• pixel labels

• **instance masks** • keypoints

• depth

• prompted region

Panoptic segmentation combines things and stuff



Things
countable objects
class + instance ID

Stuff
road, sky, vegetation
semantic region

A panoptic map gives each valid pixel a class and, for things, an instance ID.

Countable objects receive instance IDs; background regions receive semantic labels.



SAME IMAGE

— • class

• boxes

• pixel labels

• **instance masks** • keypoints

• depth

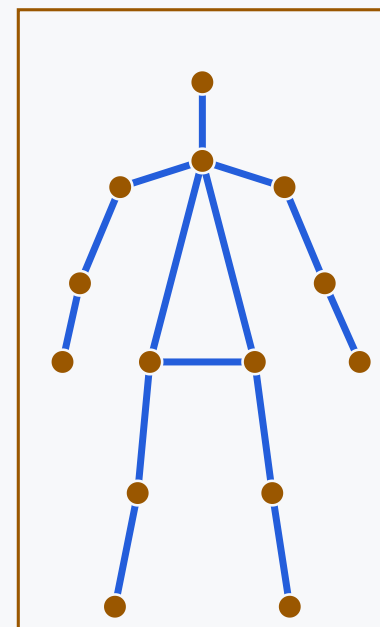
• prompted region

Other structured outputs: pose and depth

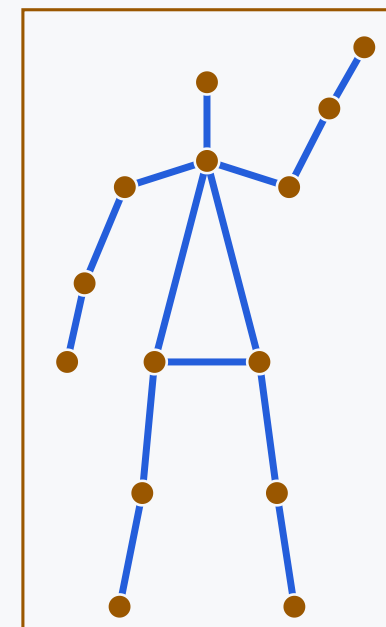


- class
- boxes
- pixel labels
- instance masks
- **keypoints**
- depth
- prompted region

A person's pose requires internal geometry



arm down



arm raised

The same box can contain different joint arrangements.

Named joints describe an arrangement that a bounding box cannot capture.



SAME IMAGE

— • class

• boxes

• pixel labels

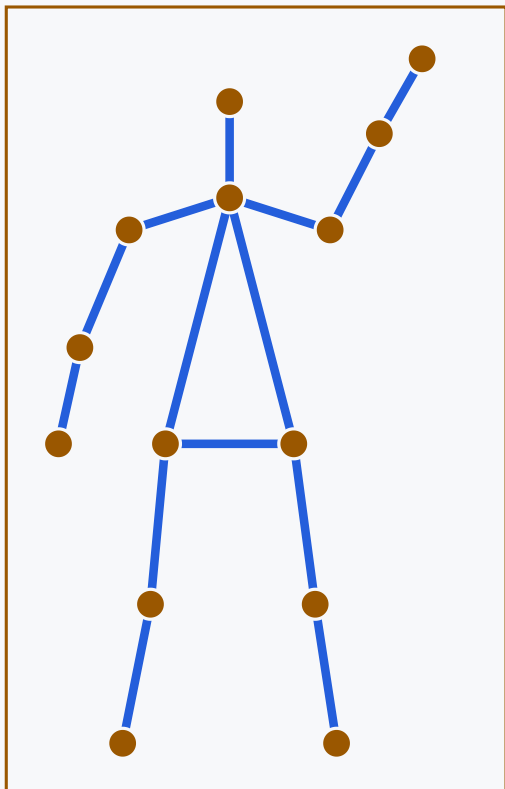
• instance masks

• **keypoints**

• depth

• prompted region

Pose can be predicted as coordinates



K joints in a defined order

1 head, 2 shoulder, ..., K ankle

Coordinates use the person crop's frame.

$$[(x_1, y_1), \dots, (x_K, y_K)] \in \mathbb{R}^{K \times 2}$$

Use validity flags for unknown joints.

Each joint has a defined identity, coordinate system, and annotation-validity flag.



SAME IMAGE

—• class

• boxes

• pixel labels

• instance masks

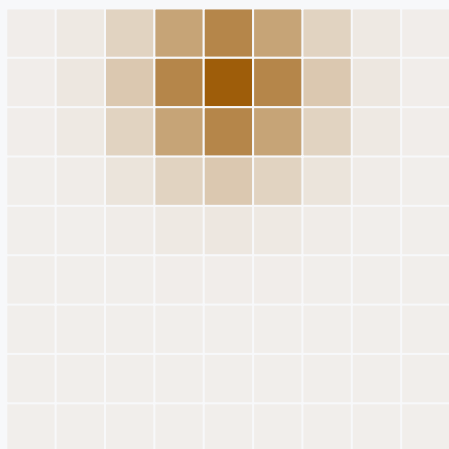
• **keypoints**

• depth

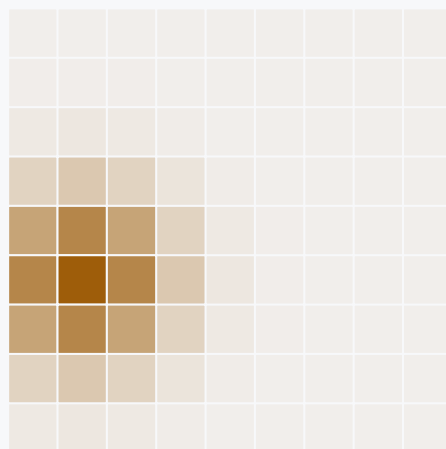
• prompted region

A heatmap represents possible joint locations

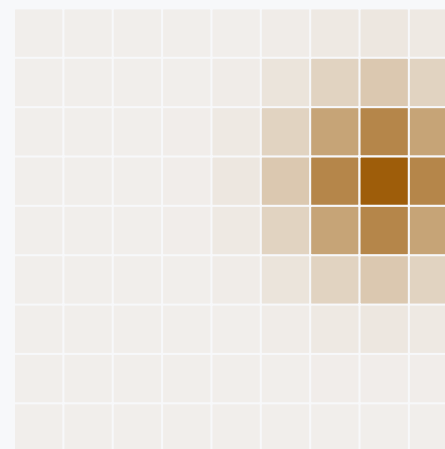
head



left wrist



right wrist



peak

 (x_k, y_k)

one person: $h \times w \times K$ heatmap scores

The output contains one spatial map for each joint type.



SAME IMAGE

—• class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• prompted region

Dense prediction can be continuous



$$D \in \mathbb{R}^{H \times W}$$

A real value at each valid spatial location.

Illustrative relative ordering: darker means nearer. Sky is invalid.

Depth assigns a number to each valid pixel instead of choosing a class.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• **depth**

• prompted region

A depth loss compares numerical values



$$\mathcal{L} = \frac{1}{|V|} \sum_{(i,j) \in V} |\hat{d}_{ij} - d_{ij}|$$

V excludes invalid measurements.

Units follow the dataset convention.

Valid depth measurements define the target.

This is dense regression, with a validity mask.

A dense regression target needs units and a valid-pixel mask.



SAME IMAGE

— • class

• boxes

• pixel labels

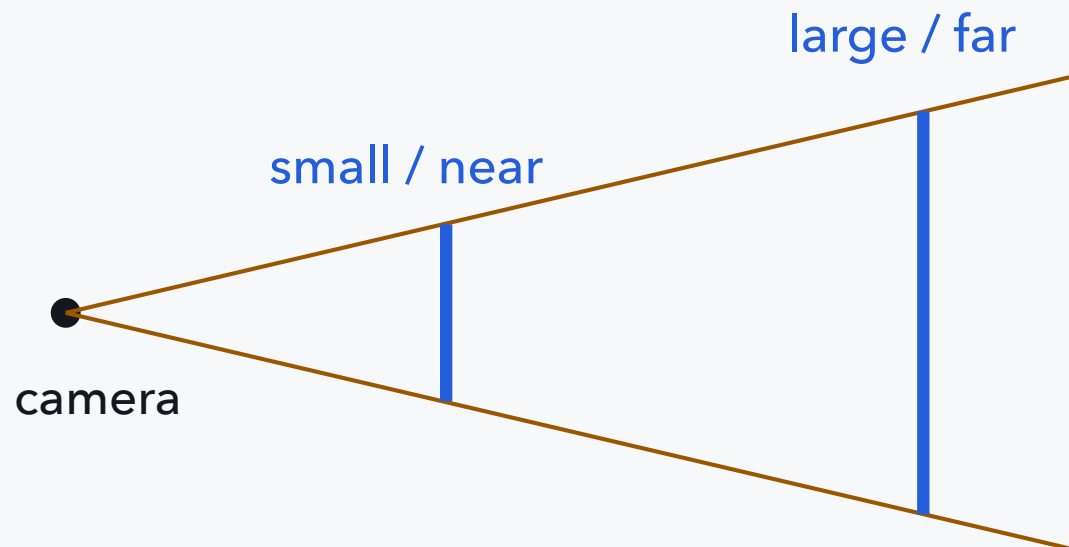
• instance masks

• keypoints

• **depth**

• prompted region

Depth values need a scale convention



Relative depth

nearer / farther ordering

absolute scale may be unknown

Metric depth

distance in a physical unit

requires a calibrated scale convention

Scale ambiguity: changing size and distance together can preserve the image.

Relative depth gives ordering; metric depth requires physical scale.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• **depth**

• prompted region

The answer can have several kinds of structure

GLOBAL

class / score

SET

boxes / object masks

DENSE

pixel labels / depth

QUERY-CONDITIONED

selected region / mask

Keypoints: a structured tuple per person; a set across people.

A query can condition which set or dense region the model returns.

| What if the output also depends on what the user is asking for?



SAME IMAGE

• class

• boxes

• pixel labels

• instance masks

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• depth

• prompted region

Now the tensor shapes have a visual meaning

TASK	TARGET MEANING	OUTPUT REPRESENTATION
classification	class ID or binary labels	C scores
detection	unordered object records	$N \times (\text{class} + \text{box})$
semantic	one class ID per pixel	$Z \in \mathbb{R}^{H \times W \times C}$
instances	one binary mask per object	$\{(c_i, b_i, M_i)\}_{i=1}^N$
keypoints	K named coordinates	$K \times 2$ per person
depth	one value per valid pixel	$D \in \mathbb{R}^{H \times W}$

A label map has $H \times W$ entries. Its classifier produces C scores at each entry.



SAME IMAGE

- class
- boxes
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- depth
- prompted region

What if the user selects the region?



- class
- boxes
- pixel labels
- instance masks
- keypoints
- depth
- **prompted region**

Which region do I want a mask for?

Which region do I want?



$$M = f(I, q)$$

Image + query produces a spatial answer.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• **prompted region**

Spatial prompts express different constraints



Positive point

include this area

Negative point

exclude this area

Box

restrict the region of interest

Previous mask

refine an earlier segmentation

A point, box, or previous mask gives the model information about the requested region.



SAME IMAGE

— • class

• boxes

• pixel labels

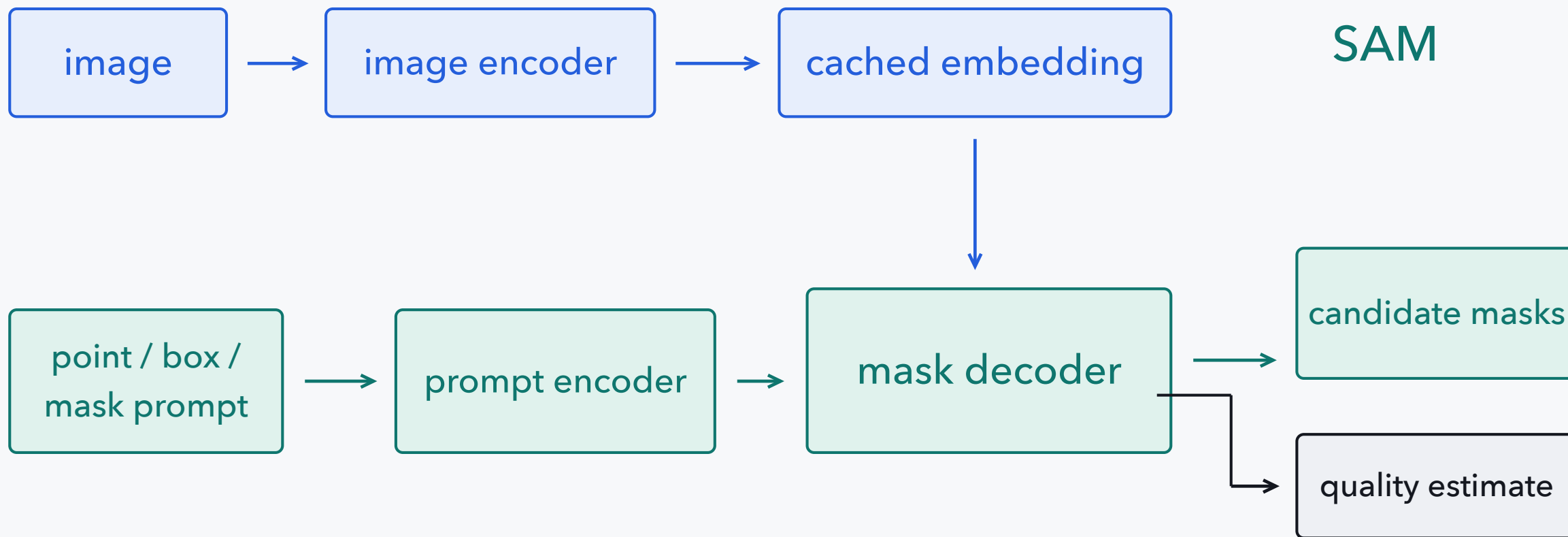
• instance masks

• keypoints

• depth

• prompted region

A promptable model combines two sources of information



A mask decoder uses visual features together with the encoded prompt.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• **prompted region**

One prompt can admit several valid masks



Does this point mean the head
or the whole dog?



part



whole object

Multiple candidate masks can express ambiguity in the prompt.



SAME IMAGE

— • class

• boxes

• pixel labels

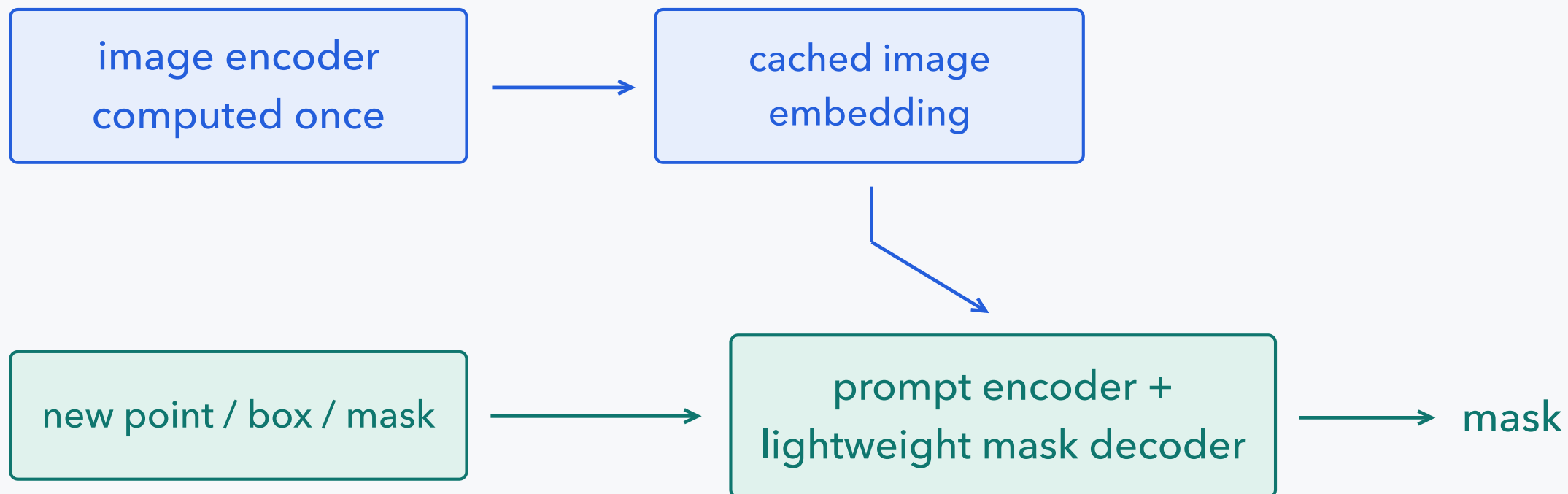
• instance masks

• keypoints

• depth

• **prompted region**

Image features can be reused across prompts



Reuse the expensive representation; change the requested region.

Interactive refinement changes the query while keeping the image representation fixed.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• **prompted region**

What if the query is language?



- class
- boxes
- pixel labels
- instance masks
- keypoints
- depth
- prompted region**

How can words select a region?



“the dog beside
the cyclist”

Words describe the referent.
Where is that referent?

SAM takes spatial prompts. A grounding stage must turn this phrase into a region.

Language describes the referent; grounding must locate it.



SAME IMAGE

— • class

• boxes

• pixel labels

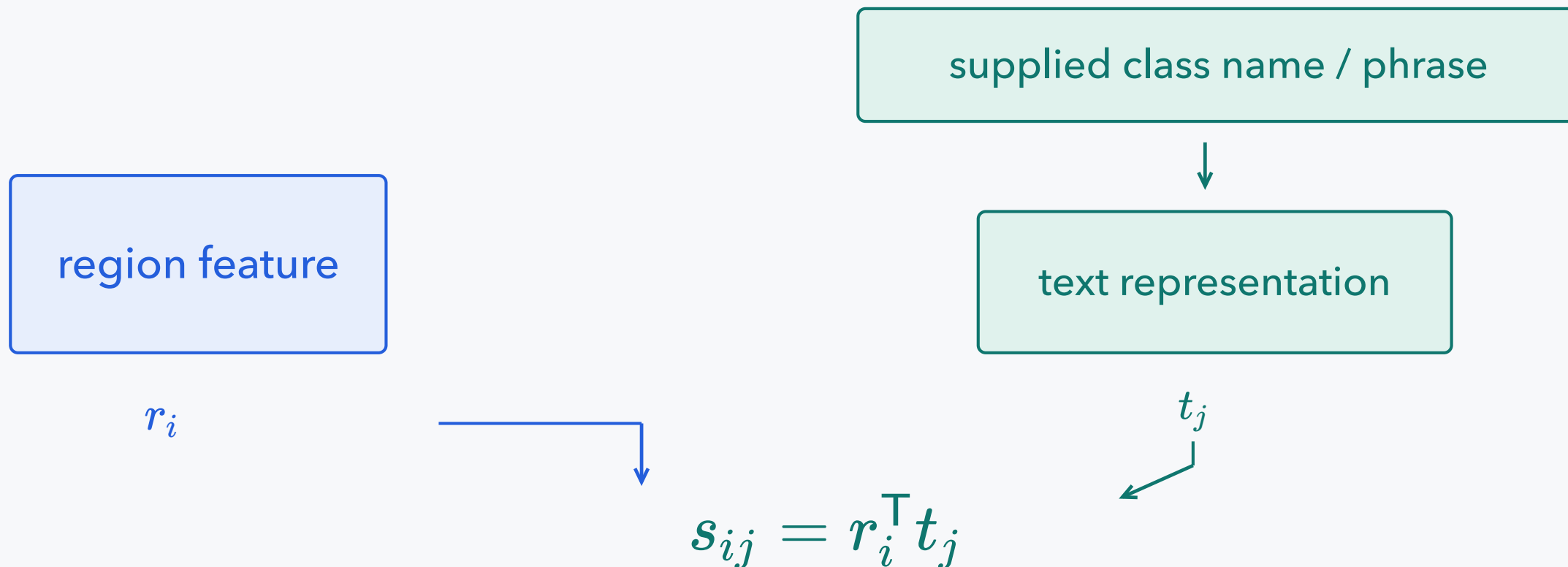
• instance masks

• keypoints

• depth

• **prompted region**

Language can supply the class representation



Comparison requires compatible learned representations.

CLIP-like comparison, now associated with spatial regions.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• **prompted region**

Phrase grounding asks which region the words refer to

"the dog beside the cyclist"



Grounding returns
a location for
the referent.

A grounded box gives a spatial referent for the phrase.



SAME IMAGE

— • class

• boxes

• pixel labels

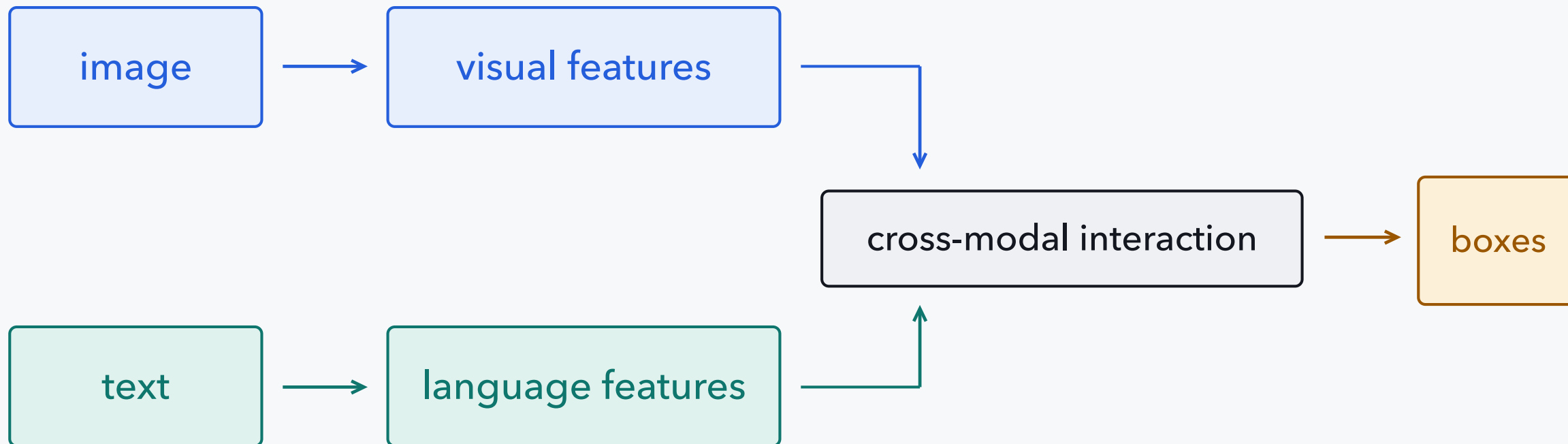
• instance masks

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• **prompted region**

Visual and language features must meet spatially



Example: a Grounding-DINO-style model

Phrase-region interaction produces text-grounded object boxes.



SAME IMAGE

— • class

• boxes

• pixel labels

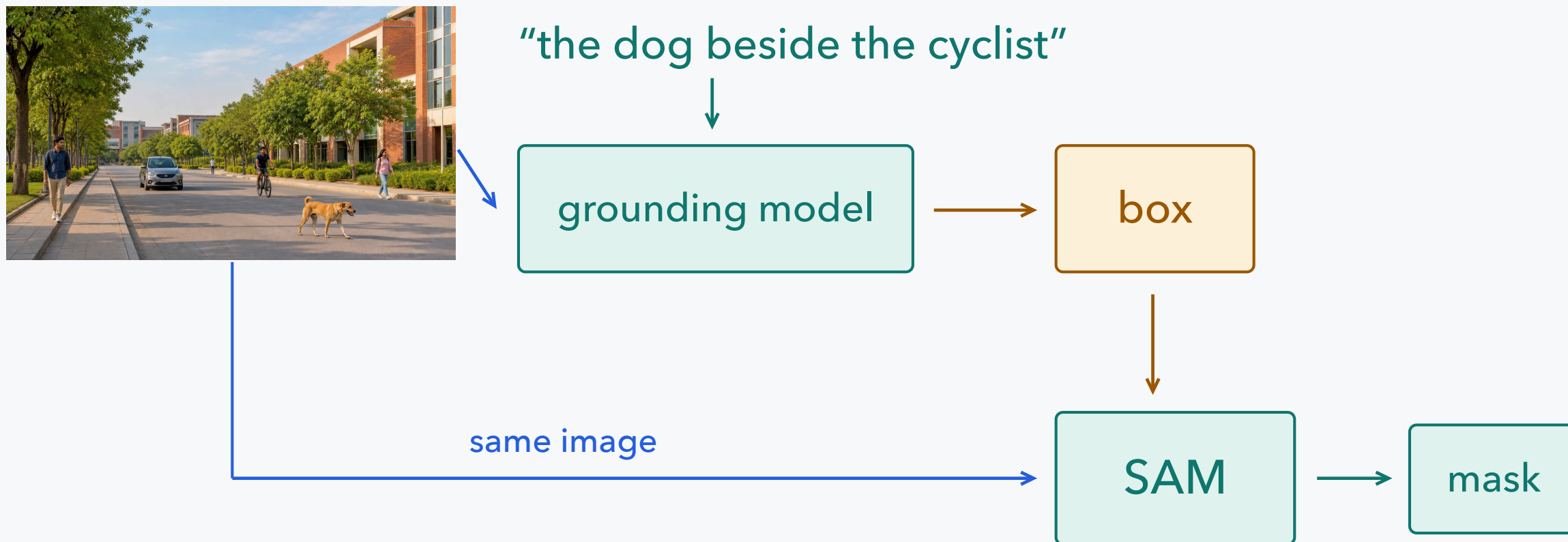
• instance masks

• keypoints

• depth

• prompted region

A grounded box becomes a spatial prompt for SAM



Language defines the target. Detection locates it. SAM refines its shape.



SAME IMAGE

— • class

• boxes

• pixel labels

• instance masks

• keypoints

• depth

• prompted region

Choose the output before the architecture



- — class
- — boxes
- — pixel labels
- — instance masks
- — keypoints
- — depth
- — prompted region

Choose the question, then the target



What is present?

class

Where are the people?

boxes / instance masks

What class is every pixel?

semantic field

What geometry is present?

keypoints / depth

Which region do I mean?

prompted mask

“the dog beside the cyclist”

grounded box, then mask

The target determines what the model must preserve.



SAME IMAGE

• class

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Which internal variables should a person understand?

We can ask language which region to look at.

Can the internal decision variables also be human-understandable?



Next: Concept Bottleneck Models



SAME IMAGE

— • class

• boxes

• pixel labels

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• **prompted region**

Optional: optical flow needs two frames

frame t

frame t + 1



at each source pixel

$$(u, v) \in \mathbb{R}^2$$

$$H \times W \times 2$$

Input: TWO frames. Output: a 2D displacement vector for each source pixel.

Occluded pixels may have no visible correspondence in the second frame.

Flow predicts a 2D displacement in the image plane for each source pixel.



SAME IMAGE

—• class

• boxes

• pixel labels

• instance masks

• keypoints

• **depth**

• prompted region

Optional: IoU and Dice on the same mask

0	1	1	0
0	1	1	0
0	1	1	0
0	0	0	0

$$\text{IoU} = \frac{TP}{TP+FP+FN} = \frac{5}{9}$$

$$\text{Dice} = \frac{2TP}{2TP+FP+FN} = \frac{10}{14}$$

Soft overlap losses use probabilities; absent classes need a convention.

Both metrics measure overlap, with different normalizations.



SAME IMAGE

—• class

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Optional: common channels-first tensor shapes

RGB image

$$B \times 3 \times H \times W$$

semantic logits

$$B \times C \times H \times W$$

semantic labels

$$B \times H \times W$$

local mask logits

$$R \times C \times m \times m$$

keypoint heatmaps

$$R \times K \times h \times w$$

depth

$$B \times 1 \times H \times W$$

Tensor layout is an implementation convention; the target meaning stays the same.



SAME IMAGE

• class

• boxes

• pixel labels

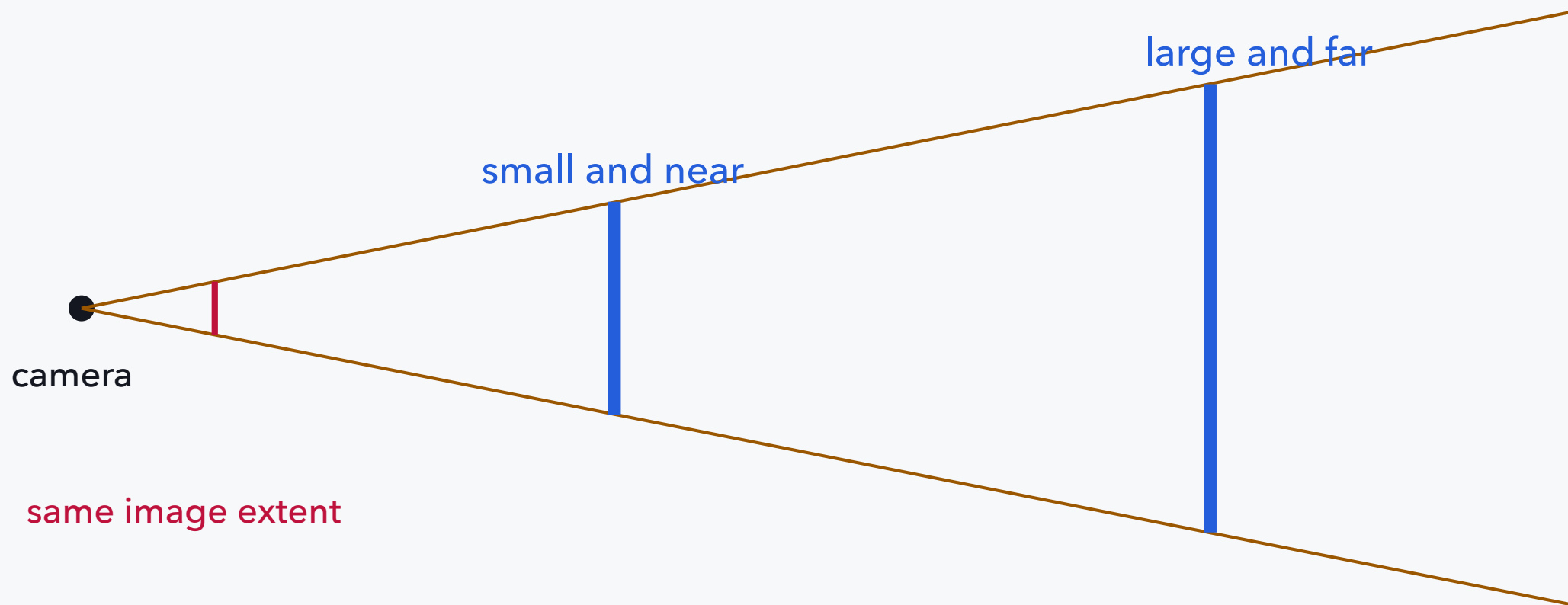
• instance masks

• keypoints

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• prompted region

A single view can leave absolute scale ambiguous



Scaling scene size and distance together can preserve the image.

Image appearance alone may be compatible with several physical scales.



SAME IMAGE

—• class

• boxes

• pixel labels

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• **depth**

• prompted region

Which target would answer each question?



How much of the image is road?

semantic segmentation

How many people are present?

detection or instance segmentation

Is the person's arm raised?

keypoints / pose

Choose the required output before choosing an architecture.



SAME IMAGE

• class

• boxes

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• prompted region

Optional: primary reading

FCN · Long, Shelhamer & Darrell (2015)

Stacked Hourglass · Newell et al. (2016)

U-Net · Ronneberger, Fischer & Brox (2015)

Depth Map Prediction · Eigen et al. (2014)

Mask R-CNN · He et al. (2017)

Robust Monocular Depth · Ranftl et al. (2022)

Panoptic Segmentation · Kirillov et al. (2019)

Segment Anything · Kirillov et al. (2023)

DeepPose · Toshev & Szegedy (2014)

RAFT · Teed & Deng (2020)

Paper links and image provenance are included in the reading notes and README.



SAME IMAGE

• class

• boxes

• pixel labels

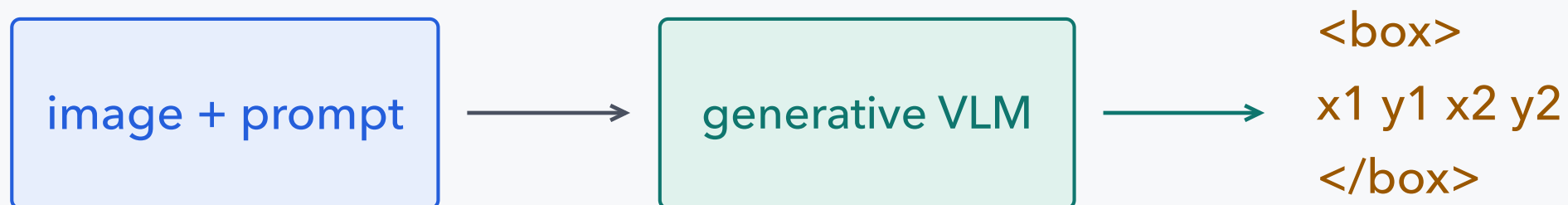
• instance masks

• keypoints

• depth

• prompted region

Optional: a VLM can generate coordinate tokens



Schematic coordinate syntax, not a universal model interface.

A valid sequence still needs spatial validation.

Generated coordinates still need a defined syntax, scale, and spatial evaluation.



SAME IMAGE

— • class

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• depth

• prompted region

Relative and metric depth answer different questions

Relative depth

nearer / farther ordering

scale may be unknown

some methods also have a shift ambiguity

Metric depth

distance in a physical unit

e.g. metres along the camera z-axis

requires a calibrated scale convention

A plausible-looking depth map does not establish metric accuracy.

The training target and evaluation protocol determine what the numbers mean.

| The meaning of depth values follows from the supervision and scale convention.



SAME IMAGE

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• prompted region